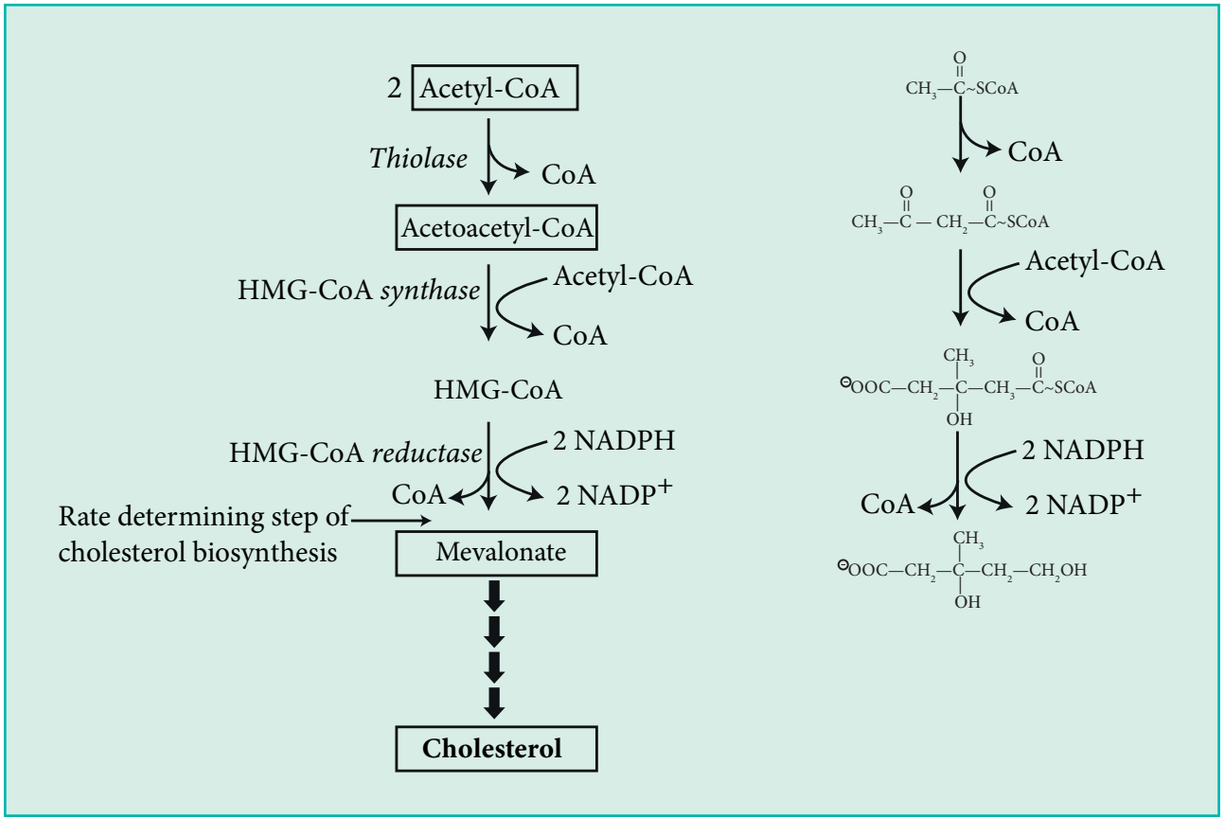
i. Formation of acetyl CoA: A molecule of acetate combines with coenzyme A (CoA) to produce Acetyl CoA in the presence of an enzyme Acetyl CoA synthetase. Acetate is the source of all carbon atoms in cholesterol.

ii. Formation of acetoacetyl CoA: Two molecules of acetyl-CoA condense to form an acetoacetyl-CoA molecule, catalyzed by the enzyme thiolase

iii. Formation of HMG CoA: The acetoacetyl-CoA further undergoes condensation with one more molecule of acetyl-CoA to form HMG-CoA (3-Hydroxy 3-Methyl Glutaryl-CoA). The enzyme which mediates this reaction is called HMG-CoA synthetase.

iv. Formation of mevalonate: The HMG-CoA is reduced to form mevalonate by NADPH dependent reductase (HMG-CoA reductase). This is the rate limiting enzyme in the pathway of cholesterol biosynthesis.



Mevalonate thus formed is then converted to squalene through various steps. Squalene, with the formation of various intermediates finally gives rise to the end-product cholesterol.

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